

1: ① $\frac{2}{5}x$ ② $\frac{1}{x}$ ③ $\frac{2}{x^2+4}$ ④ $x^2-\frac{2}{3}$
 ⑤ $\frac{1}{x}$ ⑥ $\frac{x+1}{x+2}$ 中, 属于分式 _____

2: $\frac{2x}{2-x}$ 有意义 _____

3: $\frac{x-2}{x^2-4}$ 为 0 时, $x = \underline{\hspace{2cm}}$

4: 无论 a 取何值, 下列分式总
有意义的是

A $\frac{a^2-1}{a-1}$ B $\frac{a+1}{a^2}$ C $\frac{1}{a^2+1}$ D $\frac{a}{a^2+1}$

5: 已知 $\frac{x}{3} = \frac{y}{4} = \frac{z}{5} \neq 0$, 求
 $\frac{x+y-z}{x-y+z}$ 的值.

6: <1> $(-\frac{b}{2a})^2 \cdot (\frac{3a}{b})^3 \div \frac{a^2}{4b}$

<4> $\frac{3a-2}{a^2-a} \div \frac{2a-1}{a-a^2}$

<5> $\frac{2}{m^2+m} - \frac{m+2}{m^2+2m+1}$

<6> $\frac{2x}{x+2} - \frac{x}{x-2} + \frac{4x}{x^2-4}$

<2> $\frac{2x-6}{x^2-4x+4} \div \frac{3-x}{4x^2-16} \cdot \frac{x-2}{x+3}$ 其中 $x=5$.

<7> $\frac{x-y}{x} \div (x - \frac{2xy-y^2}{x})$

<3> a, b 互为倒数, 且 $a \neq b$.

$\frac{ab^2}{a-b} + \frac{a^2b}{b-a}$

$$\langle 8 \rangle \left(\frac{1}{m+1} + \frac{1}{m-1} \right) \div \frac{m}{m^2+2m+1}$$

$$\langle 3 \rangle \left(\frac{x^2-2x}{x^2-4x+4} - \frac{3}{x-2} \right) \div \frac{x-3}{x^2-4}$$

$$\langle 9 \rangle \left(a+1 + \frac{1}{a-1} \right) \div \frac{a^3-2a^2}{a^2-4a+4}$$

$$\langle 14 \rangle \frac{a}{a+1} - \frac{a-1}{a} \div \left(\frac{a}{a+2} - \frac{1}{a^2+2a} \right)$$

$$\langle 10 \rangle \frac{x^2}{x-2} - x - 2$$

$$\langle 15 \rangle \text{ 求 } \frac{1}{1 \times 2} = 1 - \frac{1}{2}, \frac{1}{2 \times 3} = \frac{1}{2} - \frac{1}{3}$$

$$\langle 1 \rangle \frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$$

$\langle 2 \rangle$ 根据上面.

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{(x+2005)(x+2006)}$$

$$\langle 11 \rangle \frac{a^2-6a+9}{a-2} \div \left(a+2 + \frac{5}{2-a} \right)$$

$$\langle 12 \rangle \left(\frac{2x-y}{x+y} - \frac{x^2-2xy+y^2}{x^2-y^2} \right) \div \frac{x-y}{x+y}$$

$$\langle 16 \rangle a^2-2a+1=0 \text{ 求 } \frac{1}{a}$$

$$\langle 1 \rangle a + \frac{1}{a} \quad \langle 2 \rangle \left(a + \frac{1}{a} \right)^2 \quad \langle 3 \rangle a^2 + \frac{1}{a^2}$$

$$\langle 4 \rangle a^4 + \frac{1}{a^4} \quad \langle 5 \rangle \left(a - \frac{1}{a} \right)^2$$